

Distribution of Electric Charges on a Disk

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Abstract

The minimum energy configurations of N electric charges have been calculated via a procedure using simulated annealing. An overview of the initial discussion is presented. Details of the algorithm have been discussed, followed by polar plots of the configuration of charges for various values of N .

1 Introduction

In a discussion of the structure of snowflakes, ^[1] the proposition was put forward that long-range electrical forces may play a role during their formation. (We now know the cohesive forces are primarily electrostatic).

In 1985, this brought into Alexander Berezin's attention the problem of determining the least energy configuration of a system, a general problem in many situations including the formation of a snowflake. The question Berezin sought to answer was "What will be a minimum-energy configuration of a system of N equal point charges q placed inside a circle?" ^[2]

An educated guess would be to presume that the coulomb repulsion between the electric charges would cause the N charges to arrange themselves at the vertices of a regular N -sided polygon. This solution appeals due to its obvious symmetry.

Berezin argued that this configuration was only true for charges $N < 12$. An alternative configuration to preserve the total symmetry of the system would be to "expel one charge to the centre of the circle" from the circumference. For the case of $N = 12$, this configuration had a lower coulomb energy W when compared to the charges on the perimeter. Berezin proposed that this would remain true for any $N > 11$.

1.1 The 3-Dimensional Case

A small excerpt of the discussion of the 3-dimensional case is presented here for completeness but is not discussed further.

Berezin also alluded to the 3-dimensional generalisation of this problem. He suggested a “similar effect (the spontaneous “ejection” of one charge to the geometrical centre) will also be likely to happen for N charges inside the spherical surface ($N > ?$)”

The discussion was continued in a following letter. In response to Berezin’s proposition concerning the 3 dimensional problem, N.M. Queen[3] replies “it can be readily proved that for a system of point charges confined to a spherical region, the minimum energy configuration cannot conceivably have a charge at the centre. Such a configuration would contradict Earnshaw’s theorem ^A that a charge (say, at the centre of sphere) cannot be held in stable equilibrium by the electrostatic field of all the other charges.”

Martin Rees proceeds to build upon N.M Queen’s argument, explaining the motivation behind Earnshaw’s theorem.³

2 The Electrostatic Energy - W

Consider a distribution of N point charges. What would be the energy of this system? We can deduce the energy stored in the system as the work required to configure the system.^[5] Thus, we consider assembling the system an individual charge at a time, and sum the work required to place the n^{th} charge in a potential created by $n - 1$ charges. Recall for each charge Q , the work required W is $W_i = Q_i \cdot V_{i-1}(r)$, where V is the potential due to the pre-existing $n - 1$ charges.

Iterating the over the N charges with the reasoning above, we are lead to the equation :

$$W = \frac{1}{2} \cdot \frac{1}{4\pi\epsilon_0} \sum_{i=1}^N \sum_{j \neq i}^N \frac{q_i q_j}{r_{ij}} \quad (1)$$

where W is the total work required, and r_{ij} is the distance between the i^{th} and j^{th} charge. The preceding factor of two arises as in the sum we’ve counted each pair of charges twice. The lower limit $j \neq i$ in the first sum is to avoid considering the work required to place a charge in a potential created by itself.

The factor of a half also leads to computational ease as the algorithm used does not need to exclude pairs which have already been considered thus simplifying the procedure. With sufficiently small N , this does not hinder the computation time. However, for very large N , say $N > 100$, it may be beneficial to avoid recounting pairs. Such a method is not presented here.

3 The Procedure - Minimising W

In trying to determine the minimum energy configuration, we are trying to find a global optimum in a $2N$ -dimensional search space, which is why for increasing N this computation quickly becomes demanding. Within our search space, we have many local minima. Thus we use simulated annealing, which is a probabilistic approach to approximate the global minimum.

3.1 The Algorithm

The procedure starts with randomly placing N charges on a disk, by generating random polar coordinates. Polar co-ordinates were chosen as opposed to Cartesian co-ordinates as the predicted radial symmetry of the configuration becomes easy to see due to radial nature of the axis. The energy of the configuration is determined by equation (1). Note the magnitude of the point charges q and the radius of the disk R do not matter, thus the units of energy are stated as $\frac{q^2}{R}$. As the initial configuration is random, the standard deviation of the energy increases as N increases.

The algorithm proceeds by choosing a charge at random. A component of it's co-ordinate is changed by a small fixed parameter δ in either direction. The component (r or θ) and the direction are chosen at random, leading to adding or subtracting the parameter δ from the charge's position. Changing r consists of moving the charge radially inward/outward by a small length δ_r . Similarly, θ is changed by rotating the charge by δ_θ radians. δ must be chosen to be sufficiently small such that the charge is not moved over a possible minimum position, and sufficiently large such that enough of the disk is explored. Thus, we chose $\delta_r = 0.02$ and $\delta_\theta = \frac{\pi}{36}$ radians, as this lead to small changes to the configuration and lead to a reasonable computation time. The changes to the configuration were deemed small as the change in the energy for the given configuration were small. Initially, we started with $\delta_r = 0.05$ but this proved to be too large of a step size. The smaller the δ , the more accurate the calculation with the added drawback of longer computation time. The chosen δ values were sufficiently precise for the range of N that were explored. However, smaller δ would be beneficial if higher values of N are under consideration. At higher N , the configuration of charges is more sensitive as a moving a single charge changes the contribution to the energy of $N - 1$ charges, in addition to the the increased number of local minima in the $2N$ -dimensional space. Thus, smaller changes to the location of the charges is required in order to avoid skipping over minima.

The algorithm continues by computing the change in the energy ΔW due to the co-ordinate shift. $\Delta W = W(initial) - W(final)$, where the final state is the configuration of the system after a single charge has been shifted. The co-ordinate change is accepted if $\Delta W < 0$ and hence the energy W decreased.

3.2 Simulated Annealing

Following the simulated annealing procedure, the change is also accepted if the energy increased but with an associated probability. This probability is proportional to $\exp(-\Delta W/T)$ where T is another parameter, namely temperature. The decision to accept the co-ordinate change is made if $r < \exp(-\Delta W/T)$ where r is a random number r in the interval $[0,1]$. The temperature T is chosen such that initially there is a large probability that all changes are accepted and hence the system remains random.

After a number of repetitions M of randomising a charge's location and accepting/rejecting the co-ordinate change, T is exponentially reduced and the actions above are repeated another M times. The factor T is reduced by and M are parameters used to "tune" our algorithm. This procedure is repeated until changes in the configuration cease due to a reduced probability of acceptance (if $\Delta W > 0$) via a reduction of T . T is exponentially reduced, instead of say linearly, as we would like the system to gradually settle to a low energy state. This is in analogy to "cooling the system" as done in annealing. The charge has to diffuse equal to the radius of the disc at minimum, thus T must be reduced after a charge has the chance (or more precisely M chances) to do so. As the distance from the charge's initial location increases only as the square root of the number of steps in a diffusion process, a large number of steps M may be needed. The alternative is to use a larger δ . Following this procedure, the system should finalise in a low energy state.

This inclusion of the chance of changing of the charge's coordinate despite not leading to minimisation of the function, is a feature of simulated annealing. Due to this feature, we are able to avoid being stuck in local minimums whilst searching for the global optimum in the search space. However, as the initial configuration is random, and say coincidentally this configuration is close to the global minimum, the temperature may cause this configuration to deviate away from the minimum and be stuck into another local minimum. This is a probabilistic technique and thus the simulation must be repeated several times in order to confirm that the configuration is indeed the global minimum.

3.3 Tuning the Algorithm

The procedure described above was implemented with small values of N to learn how to tune the algorithm with higher values of N by extrapolation.

The 4 parameters specific to the simulated annealing procedure included:

- (i) Cooling point - this is the point at which the system is deemed to have "cooled down" and there are minimal changes to configuration and hence its energy. At this point, the global minimum should have been found. The cooling point can be decided by inspecting either the change in the energy itself and deciding it is negligible (in comparison to the energy of the system). A possible way of implementing this would be by comparing

the ratio $\Delta W/W$ to a small value, and stopping the randomising when the ratio is below the proposed value. The other option is to inspect the probability of accepting a co-ordinate change when $\Delta W > 0$. This probability tends to zero as the temperature is being reduced over cycles. Recall the probability is proportional to $\exp(-\Delta W/T)$, hence as $T \rightarrow 0$, $P(change) \rightarrow 0$, and that is only considered when $\Delta W > 0$. Hence, we can compare T to an arbitrarily small value and stop randomising when T is below this value.

- (ii) Exponentially decreasing factor for T - the important aspect of this factor is to give the configuration sufficient time at high temperatures such that initially the configuration remains random with the probability of accepting a co-ordinate change (with $\Delta W > 0$) is high. A suitable value would range anywhere from 0.75 – 0.95, but should be chosen appropriately in conjunction with the initial temperature T . A value leads to longer computation time.
- (iii) Initial temperature T - T is required such that $T \gg \Delta W$. The proposed solution is to set $T = 100 \cdot W(initial)$ where the pre-factor of 100 is arbitrarily large and $W(initial)$ is the energy of the initial configuration. This ensures that the temperature is much greater than ΔW as $\Delta W < W(initial)$.
- (iv) Number of iterations of randomising a charge at a given temperature - M . This is required to exponentially increase as N increases. It is suggested to use $N = M$ in [4], however we found this to be an insufficient number of iterations as the system did not settle in a minimum energy configuration, and the procedure seemed “incomplete”.

The annealing procedure seems insensitive to small changes in any one of these factors. However, these factors must be chosen appropriately and cohesively for varying N . Extremising any one of these factors leads to either long computation time.

3.3.1 Case Study: $N = 3$ and $N = 4$

We wanted to see how the configuration behaved with changes in factors (iii) and (iv) listed above. These two factors were investigated as amendments made to them caused greater change varying N , as discussed below.

From investigating the case $N = 3$, increasing the temperature from $T = 1, 10$ and 100 (whilst holding the other factors fixed), the results were not distinguishable. It was found that increasing T was not the best way to minimise the function quickly.

Thus we changed the number of iterations (for each temperature held constant) and for small incremental changes (i.e. increasing the number of iterations) the charged settled into the minimum configuration quicker.

For the case of $N = 4$, a more systematic approach was used to investigate the behaviour. A combination of low/high temperature T in conjunction with small/large number of iterations M lead to the following conclusions:

- Low T and small M lead to greater standard deviation in the energy of the configuration W . The configuration was not at a minimum
- High T and small M lead to a reduction of the standard deviation but the mean of the energies W showed negligible changed
- Low T and large M lead to a mean W closer to than of minimum energy. The standard deviation of W however was similar to the low T and small M case.
- High T and large M lead to the energy being minimised and the standard deviation being reduced significantly

The above were verified with $N=5,6,7,8$ with larger different range of temperatures and values of M . We concluded for quicker minimisation a large number of iterations M is required, along with a sufficiently high temperature. However, increasing the value of M lead to more noticeable change in the procedure than altering T .

4 Configurations for higher N

Polar plot of the supposed minimum energy configurations have been plotted below for $N=11, 12, 16, 17, 20, 21$ and 50 .

We were unable to verify the result for $N = 11$, and had inconsistencies in our results for larger N . This was due to lack of computational power, and thus we weren't able to increase M high enough in order for the computation to complete in a minimum configuration.

For the case of $N = 11$, we arrived at two different configuration both different from the proposed charges at the perimeter. We suspect this is due to the initial random placing of the charges on the disk in addition to randomly choosing a charge out of the configuration to shift. As opposed to following a procedure in which each charge is definitely considered in turn and is moved. Within the simulated annealing procedure, we this is hard to escape. Similar results were found for the case $N = 10$ but it not presented here.

With the $N = 12$ case, we were able to verify the minimum energy configuration with one charged being expelled to the middle. The simulation found another local minima with two charges in the middle of the disk.

For the cases of $N = 16$ and $N = 17$, (Figure 1) the time allocated to the algorithm was insufficient, resulting in the configuration approaching the minimum. Again, two different local minima are found for each. The charges seem to be approaching a symmetrical configuration, k -sided polygons being formed in the centre of the circle with k "expelled" charges. However, more accurate numerical results are required if we are to verify the proposed symmetry of the configurations. A similar argument holds for our results of $N = 20$ and $N = 21$.

Figure 4 shows a plot of the minimum energy configuration of $N = 50$ charges. The configuration seems to be approaching two concentric circles with one charge expelled in the middle. Insufficient run time lead to the charges not reaching their minimum energy configuration.

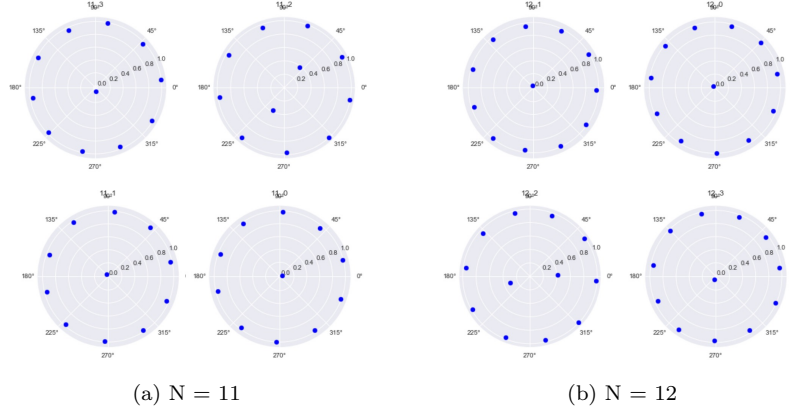


Figure 1: Polar plots of the configuration of N charges for $N = 11$ and $N = 12$

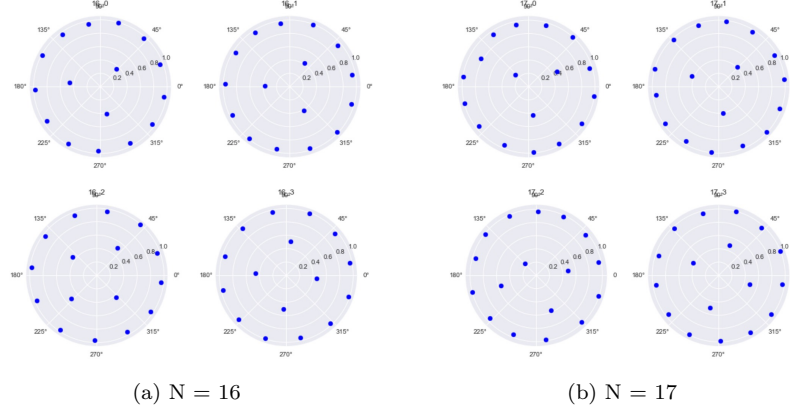


Figure 2: Polar plots of the configuration of N charges for $N = 16$ and $N = 17$

5 Remarks

The simulated annealing procedure has use been to find minimum energy configuration of N charges on a disk with limited success. The procedure does not scale well with increasing values of N as the number of local minima increases. The computation requires exponentially increasing time with increasing N . It is difficult to tell if a given configuration is symmetric due to lack of precision of the results. A more systematic approach is required for initialising original co-ordinates, as even with sustained high temperatures, configurations are likely to be stuck in local minima.

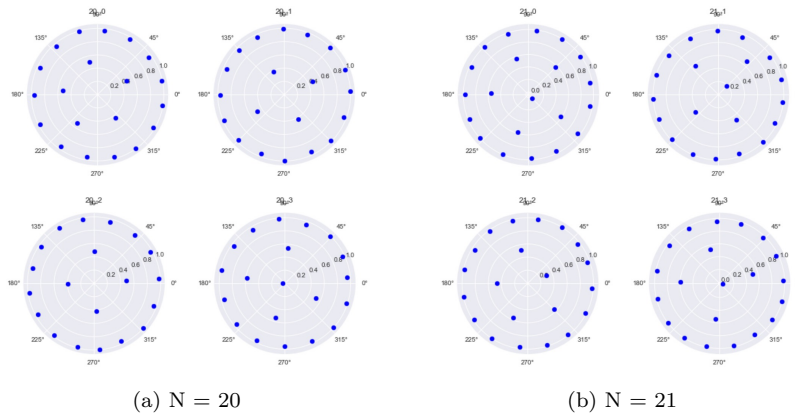


Figure 3: Polar plots of the configuration of N charges for $N = 20$ and $N = 21$

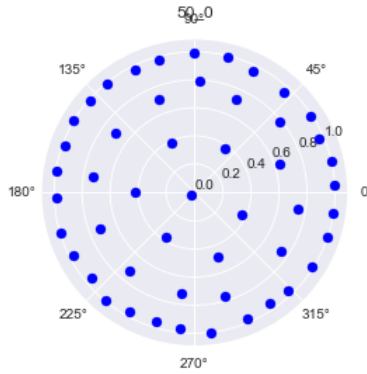


Figure 4: Polar plot of the minimum energy configuration of $N = 50$ charges

6 References

1. No pattern yet for snowflakes – W.S. Mortley. (Nature 313, 638; Feb 1985)
2. An unexpected result in classical electrostatics – Alexander A Berezin (Nature 315, 104; May 1985)
3. The distribution of charges in classical electrostatics – M Queen, Martin Rees and Berezin (Nature 317, 208; Sept 1985)
4. Electrostatic energy minimisation by simulated annealing – L.T.Willie and J.Vennik (Journal of Physics A: Mathematical and General Vol. 18, L1113)
5. Introduction to Electrodynamics - David J. Griffiths

7 Appendix

A - Earnshaw's Theorem

Earnshaw's theorem states that a collection of point charges cannot be maintained in a stable stationary equilibrium configuration solely by the electrostatic interaction of the charges.

The case of a point charge in an static electric field is a consequence of Gauss's law. For a particle to be in a stable equilibrium, small perturbations to the particle's position should not break the equilibrium and the particle should return to its previous position. Gauss's Law says that the divergence of the electric force field is zero in free space. An electrical force $\mathbf{F}(\mathbf{r})$ from a potential $U(\mathbf{r})$ will always satisfy Laplace's equation and hence be divergence free:

$$\nabla \cdot \mathbf{F} = \nabla \cdot (-\nabla U) = -\nabla^2 U = 0.$$

Therefore, there are no local minima or maxima of the field potential in free space, only saddle points. A stable equilibrium of the particle cannot exist and there must be an instability in some direction.